

# RINNS OF ISLAY LH, United Kingdom

WMO: 031020

Lat: 55.673N

Lon: 6.513W

Elev: 75

StdP: 14.66

Time Zone: 0.00 (EUW)

Period: 86-98

WBAN: 99999

## Annual Heating, Humidification, and Ventilation Design Conditions

| Coldest Month | Heating DB |      | Humidification DP/MCDB and HR |      |      |      |      |      | Coldest Month WS/MCDB |      |      |      | MCWS/PCWD to 99.6% DB |      | WSF   |
|---------------|------------|------|-------------------------------|------|------|------|------|------|-----------------------|------|------|------|-----------------------|------|-------|
|               |            |      | 99.6%                         |      |      | 99%  |      |      | 0.4%                  |      | 1%   |      |                       |      |       |
|               | 99.6%      | 99%  | DP                            | HR   | MCDB | DP   | HR   | MCDB | WS                    | MCDB | WS   | MCDB | MCWS                  | PCWD |       |
| (a)           | (b)        | (c)  | (d)                           | (e)  | (f)  | (g)  | (h)  | (i)  | (j)                   | (k)  | (l)  | (m)  | (n)                   | (o)  | (p)   |
| 2             | 32.8       | 34.8 | 24.5                          | 18.7 | 36.6 | 26.7 | 20.7 | 37.1 | 55.4                  | 41.9 | 51.2 | 43.2 | 13.2                  | 50   | 0.885 |

## Annual Cooling, Dehumidification, and Enthalpy Design Conditions

| Hottest Month | Hottest Month DB Range | Cooling DB/MCWB |      |      |      |      |      | Evaporation WB/MCDB |      |      |      |      |      | MCWS/PCWD to 0.4% DB |      |
|---------------|------------------------|-----------------|------|------|------|------|------|---------------------|------|------|------|------|------|----------------------|------|
|               |                        | 0.4%            |      | 1%   |      | 2%   |      | 0.4%                |      | 1%   |      | 2%   |      |                      |      |
|               |                        | DB              | MCWB | DB   | MCWB | DB   | MCWB | WB                  | MCDB | WB   | MCDB | WB   | MCDB | MCWS                 | PCWD |
| (a)           | (b)                    | (c)             | (d)  | (e)  | (f)  | (g)  | (h)  | (i)                 | (j)  | (k)  | (l)  | (m)  | (n)  | (o)                  | (p)  |
| 8             | 5.3                    | 64.6            | 59.0 | 62.1 | 57.8 | 60.3 | 57.0 | 60.3                | 63.1 | 59.0 | 60.9 | 57.9 | 59.6 | 12.0                 | 90   |

| Dehumidification DP/MCDB and HR |      |      |      |      |      |      |      |      | Enthalpy/MCDB |      |      |      |      |      | Extreme<br>Max WB |
|---------------------------------|------|------|------|------|------|------|------|------|---------------|------|------|------|------|------|-------------------|
| 0.4%                            |      |      | 1%   |      |      | 2%   |      |      | 0.4%          |      | 1%   |      | 2%   |      |                   |
| DP                              | HR   | MCDB | DP   | HR   | MCDB | DP   | HR   | MCDB | Enth          | MCDB | Enth | MCDB | Enth | MCDB |                   |
| (a)                             | (b)  | (c)  | (d)  | (e)  | (f)  | (g)  | (h)  | (i)  | (j)           | (k)  | (l)  | (m)  | (n)  | (o)  |                   |
| 59.1                            | 75.0 | 61.4 | 58.0 | 72.1 | 60.1 | 57.0 | 69.5 | 58.9 | 26.6          | 63.4 | 25.7 | 60.9 | 25.0 | 59.8 | 67.3              |

## Extreme Annual Design Conditions

| Extreme Annual WS |      |      | Extreme Annual Temperature |      |                    |     | n-Year Return Period Values of Extreme Temperature |      |            |      |            |      |            |      |    |    |
|-------------------|------|------|----------------------------|------|--------------------|-----|----------------------------------------------------|------|------------|------|------------|------|------------|------|----|----|
|                   |      |      | Mean                       |      | Standard Deviation |     | n=5 years                                          |      | n=10 years |      | n=20 years |      | n=50 years |      |    |    |
| 1%                | 2.5% | 5%   | Min                        | Max  | Min                | Max | Min                                                | Max  | Min        | Max  | Min        | Max  | Min        | Max  | DB | WB |
| (a)               | (b)  | (c)  | (e)                        | (f)  | (g)                | (h) | (i)                                                | (j)  | (k)        | (l)  | (m)        | (n)  | (o)        | (p)  |    |    |
| (4) 46.5          | 41.3 | 35.5 | 29.1                       | 68.0 | 3.0                | 3.9 | 26.9                                               | 70.9 | 25.2       | 73.2 | 23.5       | 75.4 | 21.3       | 78.2 |    |    |
| (5)               |      |      | 27.4                       | 63.0 | 2.9                | 2.0 | 25.3                                               | 64.4 | 23.6       | 65.6 | 22.0       | 66.7 | 19.9       | 68.2 |    |    |

## Monthly Climatic Design Conditions

|                                                                        |             | Annual | Jan   | Feb   | Mar   | Apr   | May   | Jun   | Jul   | Aug   | Sep   | Oct   | Nov   | Dec   |
|------------------------------------------------------------------------|-------------|--------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|
|                                                                        |             | (d)    | (e)   | (f)   | (g)   | (h)   | (i)   | (j)   | (k)   | (l)   | (m)   | (n)   | (o)   | (p)   |
| (6) Temperatures, Degree-Days and Degree-Hours                         | DBAvg       | 49.0   | 43.3  | 42.1  | 43.6  | 45.3  | 49.5  | 52.9  | 55.9  | 56.7  | 54.6  | 51.4  | 47.5  | 44.6  |
|                                                                        | DBStd       | 6.03   | 3.89  | 3.91  | 3.14  | 2.93  | 3.68  | 3.05  | 2.58  | 2.49  | 2.63  | 3.56  | 3.78  | 4.05  |
|                                                                        | HDD50       | 1115   | 208   | 220   | 200   | 141   | 52    | 6     | 0     | 0     | 2     | 28    | 89    | 169   |
|                                                                        | HDD65       | 5846   | 672   | 640   | 665   | 590   | 482   | 365   | 283   | 259   | 311   | 423   | 524   | 632   |
|                                                                        | CDD50       | 746    | 1     | 0     | 0     | 1     | 35    | 91    | 183   | 206   | 141   | 70    | 15    | 3     |
|                                                                        | CDD65       | 1      | 0     | 0     | 0     | 0     | 0     | 1     | 0     | 0     | 0     | 0     | 0     | 0     |
|                                                                        | CDH74       | 0      | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     |
|                                                                        | CDH80       | 0      | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     |
|                                                                        | WSAvg       | 18.3   | 23.1  | 23.6  | 21.2  | 16.0  | 14.8  | 13.0  | 13.0  | 13.4  | 16.8  | 19.9  | 21.8  | 22.8  |
|                                                                        | PrecAvg     | 47.8   | 4.9   | 3.8   | 3.5   | 3.0   | 2.8   | 3.2   | 3.3   | 3.7   | 3.9   | 5.2   | 5.2   | 5.2   |
| (15) Precipitation                                                     | PrecMax     | 58.7   | 7.8   | 8.1   | 7.3   | 5.3   | 5.4   | 6.2   | 5.6   | 7.9   | 7.6   | 10.0  | 9.0   | 10.3  |
|                                                                        | PrecMin     | 36.5   | 1.1   | 1.4   | 1.3   | 1.3   | 0.7   | 1.1   | 1.2   | 0.7   | 0.7   | 1.5   | 2.6   | 2.4   |
|                                                                        | PrecStd     | 5.5    | 1.6   | 1.7   | 1.4   | 1.1   | 1.3   | 1.3   | 1.1   | 1.6   | 1.5   | 2.1   | 1.5   | 1.9   |
|                                                                        |             |        |       |       |       |       |       |       |       |       |       |       |       |       |
| (19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures | 0.4%        | DB     | 52.3  | 51.6  | 50.8  | 54.7  | 64.8  | 67.5  | 69.3  | 67.4  | 63.0  | 59.8  | 56.5  | 53.9  |
|                                                                        |             | MCWB   | 50.3  | 50.4  | 48.9  | 49.5  | 56.3  | 60.1  | 62.6  | 61.5  | 58.8  | 57.2  | 54.6  | 52.6  |
|                                                                        | 2%          | DB     | 50.9  | 49.7  | 49.5  | 51.9  | 60.2  | 62.4  | 63.7  | 64.0  | 60.6  | 58.2  | 54.9  | 52.3  |
|                                                                        |             | MCWB   | 49.5  | 48.4  | 47.8  | 48.7  | 53.7  | 56.8  | 59.1  | 59.7  | 57.2  | 56.0  | 53.2  | 51.1  |
|                                                                        | 5%          | DB     | 49.8  | 48.4  | 48.6  | 50.4  | 56.3  | 59.3  | 61.2  | 61.6  | 59.3  | 57.1  | 53.8  | 51.1  |
|                                                                        |             | MCWB   | 48.5  | 46.9  | 47.1  | 48.0  | 51.7  | 54.9  | 57.7  | 58.3  | 56.2  | 55.0  | 52.1  | 49.8  |
|                                                                        | 10%         | DB     | 48.6  | 47.2  | 47.8  | 49.3  | 54.1  | 56.8  | 59.4  | 60.0  | 58.3  | 56.0  | 52.6  | 50.1  |
|                                                                        |             | MCWB   | 47.3  | 45.7  | 46.4  | 47.4  | 50.8  | 53.5  | 56.5  | 57.5  | 55.5  | 54.1  | 50.8  | 48.8  |
|                                                                        | 0.4%        | WB     | 51.1  | 50.8  | 50.0  | 50.9  | 57.4  | 60.4  | 63.4  | 62.2  | 60.2  | 58.1  | 55.5  | 53.4  |
|                                                                        |             | MCDB   | 51.6  | 51.1  | 50.6  | 52.5  | 64.2  | 66.2  | 68.7  | 65.9  | 62.1  | 59.3  | 56.3  | 53.6  |
| (27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures | 2%          | WB     | 50.0  | 48.9  | 48.6  | 49.6  | 54.2  | 57.5  | 60.2  | 60.3  | 58.5  | 56.7  | 53.7  | 51.3  |
|                                                                        |             | MCDB   | 50.7  | 49.4  | 49.2  | 50.9  | 58.2  | 61.6  | 63.0  | 62.7  | 59.7  | 57.6  | 54.4  | 51.8  |
|                                                                        | 5%          | WB     | 48.8  | 47.4  | 47.8  | 48.7  | 52.7  | 55.6  | 58.4  | 59.2  | 57.3  | 55.7  | 52.5  | 50.2  |
|                                                                        |             | MCDB   | 49.6  | 48.1  | 48.4  | 49.8  | 55.7  | 58.3  | 60.3  | 60.8  | 58.4  | 56.6  | 53.5  | 50.9  |
|                                                                        | 10%         | WB     | 47.6  | 46.2  | 46.8  | 47.8  | 51.3  | 54.2  | 57.1  | 58.2  | 56.4  | 54.6  | 51.3  | 49.1  |
|                                                                        |             | MCDB   | 48.5  | 47.1  | 47.5  | 48.9  | 53.6  | 56.3  | 58.9  | 59.6  | 57.7  | 55.8  | 52.4  | 50.0  |
|                                                                        | 5% DB       | MDBR   | 5.0   | 5.1   | 5.1   | 5.8   | 6.8   | 6.2   | 5.6   | 5.3   | 5.5   | 5.0   | 4.7   | 4.9   |
|                                                                        |             | MCDBR  | 5.3   | 5.3   | 5.1   | 6.7   | 9.6   | 10.1  | 8.6   | 7.4   | 6.6   | 5.7   | 4.7   | 4.8   |
|                                                                        |             | MCWBR  | 5.8   | 5.9   | 4.9   | 5.2   | 6.4   | 6.8   | 5.7   | 4.9   | 5.3   | 5.8   | 5.5   | 5.7   |
|                                                                        |             | MCWBR  | 5.3   | 5.1   | 4.9   | 5.8   | 8.3   | 8.2   | 7.9   | 6.3   | 5.6   | 5.2   | 4.7   | 4.9   |
| (35) Mean Daily Temperature Range                                      | 5% WB       | MCWBR  | 5.8   | 6.0   | 5.0   | 5.1   | 6.2   | 6.2   | 5.8   | 4.7   | 5.6   | 5.9   | 5.7   | 6.0   |
|                                                                        |             |        |       |       |       |       |       |       |       |       |       |       |       |       |
|                                                                        |             |        |       |       |       |       |       |       |       |       |       |       |       |       |
|                                                                        |             |        |       |       |       |       |       |       |       |       |       |       |       |       |
| (40) Clear-Sky Solar Irradiance                                        | taub        |        | 0.344 | 0.353 | 0.372 | 0.383 | 0.387 | 0.378 | 0.381 | 0.367 | 0.365 | 0.349 | 0.334 | 0.331 |
|                                                                        | taud        |        | 2.198 | 2.292 | 2.311 | 2.328 | 2.367 | 2.430 | 2.428 | 2.479 | 2.468 | 2.417 | 2.272 | 2.171 |
|                                                                        | Ebn at Noon |        | 170   | 223   | 250   | 266   | 271   | 275   | 272   | 269   | 252   | 224   | 175   | 143   |
|                                                                        | Edh at Noon |        | 19    | 25    | 31    | 35    | 36    | 34    | 33    | 30    | 26    | 22    | 17    | 16    |
| (44) All-Sky Solar Radiation                                           | RadAvg      |        | 150   | 367   | 723   | 1223  | 1583  | 1613  | 1498  | 1210  | 812   | 441   | 190   | 106   |
|                                                                        | RadStd      |        | 16    | 33    | 54    | 78    | 133   | 153   | 100   | 95    | 48    | 40    | 17    | 10    |

## Historical Trends

|                              | DBAvg | Heating |     |        | Cooling |       |       | Degree-Days |       |       |       |
|------------------------------|-------|---------|-----|--------|---------|-------|-------|-------------|-------|-------|-------|
|                              |       | 99% DB  |     | 99% DP | 1% DB   | 1% WB | 1% DP | HDD50       | HDD65 | CDD50 | CDD65 |
|                              |       | (d)     | (e) | (f)    | (g)     | (h)   | (i)   | (j)         | (k)   | (l)   | (m)   |
| (46) Station Only            |       | N/A     | N/A | N/A    | N/A     | N/A   | N/A   | N/A         | N/A   | N/A   | N/A   |
| (47) Regional (38 neighbors) |       | N/A     | N/A | N/A    | N/A     | N/A   | N/A   | N/A         | N/A   | N/A   | N/A   |

Nomenclature: See separate page